

Slope is the ratio that is used to measure steepness. Without realizing it, you encounter situations that involve slope every day.

Think about the simple experience of walking up or down a staircase. If the steps are built according to established building codes, you hardly need to think as you climb the stairs. If the building codes aren't followed, you may very well trip or fall.

A stair has two parts: the *rise* and the *tread*. The minimum rise is 5 inches, and the minimum tread is 9 inches. Builders use this formula: $2 \times \text{Rise} + \text{Tread} = 24.6$ inches



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A builder wants to use a rise of 5.75 inches. Find the length of the tread. Find the slope of the staircase.

Step 1

Use the formula $2 \times \text{Rise} + \text{Tread} = 24.6 \text{ in.}$

$$2 \times \text{Rise} + \text{Tread} = 24.6 \text{ in}$$

$$2 \times 5.75 + T = 24.6 \text{ in}$$

$$11.5 + T = 24.6 \text{ in}$$

$T = 13.1 \text{ in}$

Step 2

Find the slope. Substitute tread for run in the slope ratio.

$$\frac{\text{rise}}{\text{run}} = \frac{5.75}{13.1} \approx 0.44$$

For a rise of 5.75 inches, the tread should be **13.1 inches**.

The slope is about **0.44**.

Finding the steepness of a ramp is another common application of slope. Disability access codes state that the

maximum slope for a ramp is 1:12 (or $\frac{1}{12}$). However, slopes between 1:16 and 1:20 are preferred.

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One Stop Office Supplies needs to build an access ramp to get to the store's entrance. The ramp needs to rise a distance of 3 feet. If the owner wants to use a 1:16 slope for the ramp, how many feet long will the ramp need to be?

Write and solve a proportion: $\frac{\text{rise}}{\text{run}}$

$$\frac{1}{16} = \frac{3}{x}$$

$$x = 3 \cdot 16$$

$x = 48 \text{ ft}$

The ramp needs to be **48 feet** long.

Slope is a ratio. Besides using a colon, you can also write slope as a fraction or a decimal. You can easily compare slopes as decimals on a calculator.

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A ramp rises 3 inches for every 20 inches of length. Does the ramp exceed the building limit of a 1:12 slope?

Turn the slopes into decimals: $\frac{1}{12} = 0.0833$ $\frac{3}{20} = 0.15$

Yes, 0.15 is greater than 0.08, so the slope of the ramp exceeds the building limit.

Key Terms

 negative reciprocal

 negative slope

 parallel

 perpendicular

 positive slope